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RESEARCH ARTICLE

# Optimizing Secure Multi-User ISAC Systems With STAR-RIS: A Deep Reinforcement Learning Approach for 6G Networks

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**ABSTRACT** The rapid evolution of wireless communication technologies and the increasing demand for multi-functional systems have led to the emergence of integrated sensing and communication (ISAC) as a key enabler for future 6G networks. Simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) have recently garnered significant attention for their ability to enhance signal coverage and improve system efficiency. This paper investigates a STAR-RIS-assisted ISAC system designed to secure communication for multiple legitimate users (LUs) while safeguarding against multiple eavesdroppers (Eves). By jointly optimizing the base station (BS) transmit beamforming, STAR-RIS transmission and reflection coefficients, and receive filters, the proposed framework aims to maximize the long-term average secrecy rate for all LUs. Constraints are imposed to ensure minimum echo signal-to-noise ratios (SNRs) for sensing and meet the achievable rate requirements of LUs. To address the inherent complexity of this non-convex problem, two deep reinforcement learning (DRL) algorithms are proposed. Numerical results demonstrate that the proposed system achieves significant improvements in secrecy rate compared to conventional RIS setups. This work provides a scalable and efficient approach for secure multi-user ISAC systems, making it highly relevant for future 6G networks, smart cities, and IoT applications.

**INDEX TERMS** Integrated sensing and communication (ISAC), deep reinforcement learning (DRL), STAR-RIS, secure communication, multi-user secrecy rate, 6G, multiple eavesdroppers.

## I. INTRODUCTION

The next-generation wireless communication networks, often referred to as sixth-generation (6G) networks, aim to revolutionize connectivity by integrating high-performance communication and precise sensing functionalities. Unlike previous generations, 6G envisions a unified framework for communication and sensing, termed ISAC, to address emerging demands in autonomous driving, smart cities, healthcare, and industrial automation [1], [2]. By combining these traditionally distinct functions, ISAC systems can simultaneously enhance spectral efficiency, reduce hardware complexity, and enable novel applications that were previously unattainable [3].

A key challenge for ISAC systems is their dependency on the propagation environment, which is often unpredictable

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due to dynamic obstacles, mobility, and complex terrains. Conventional solutions, such as adaptive beamforming or relay systems, often fall short in addressing these dynamic challenges [4]. RIS have emerged as a transformative technology capable of mitigating these issues by dynamically controlling the wireless channel environment. RIS consists of programmable meta-materials that manipulate electromagnetic waves to enhance signal quality, reduce interference, and extend coverage [5]. However, traditional RIS is limited in its ability to address multi-user scenarios, as it relies solely on reflective mechanisms, confining coverage to users positioned on the same side of the surface as the base station (BS) [6].

To overcome the inherent limitations of traditional RIS, STAR-RIS technology has been introduced. Unlike conventional RIS, STAR-RIS can both transmit and reflect signals, enabling 360-degree coverage. This feature makes STAR-RIS particularly suitable for ISAC systems, where both communication and sensing functions often require diverse and adaptive coverage zones [7]. STAR-RIS achieves this functionality by dividing its elements into transmission and reflection modes, offering greater flexibility in beamforming and resource allocation [8]. Recent studies have demonstrated the potential of STAR-RIS in enhancing signal quality, improving sensing accuracy, and enabling secure communication [9].

In addition to communication and sensing performance, security remains a critical concern for ISAC systems, particularly in scenarios involving multiple legitimate users and multiple eavesdroppers. Eavesdroppers pose significant risks by intercepting legitimate signals, undermining the integrity and confidentiality of communications. Traditional security techniques, such as cryptographic encryption, are often insufficient in high-mobility or low-latency environments, where PLS approaches are more suitable [10]. STAR-RIS technology has shown promise in enhancing PLS by dynamically controlling signal propagation to maximize secrecy rates and minimize information leakage [11].

Physical Layer Security

Optimizing the performance of STAR-RIS-assisted ISAC systems is a highly complex task, involving the joint design of BS transmit beamforming, STAR-RIS transmission and reflection coefficients, and receive filters. This multi-variable optimization problem is further complicated by the dynamic and non-convex nature of wireless channels. Traditional optimization methods, which rely on convex approximations, often fail to achieve real-time adaptability and global optimality in such scenarios [12]. Advanced machine learning techniques, particularly DRL, offer a promising alternative. DRL algorithms can efficiently explore high-dimensional solution spaces and adapt to dynamic environments, making them ideal for optimizing STAR-RIS-assisted ISAC systems [13].

BS Base Station

Deep Reinforcement Learning

This paper leverages DRL techniques to jointly optimize critical system parameters in a STAR-RIS-assisted ISAC framework, ensuring secure and efficient multi-user communication. The proposed framework addresses the challenges of multi-user and multi-eavesdropper scenarios by employing two advanced DRL algorithms: DDPG and SAC. These algorithms are designed to maximize the long-term average secrecy rate of LUs while ensuring robust sensing performance under dynamic channel conditions [14].

legitimate user

The proposed system is expected to play a pivotal role in the evolution of 6G networks, enabling innovative applications such as secure IoT, autonomous vehicle communication, and resilient smart city infrastructure. By addressing the dual challenges of communication and sensing performance and ensuring robust security, this work sets the stage for scalable and adaptive solutions in next-generation wireless networks [15].

### A. RELATED WORK

The development of RIS and their application to ISAC systems have attracted considerable research attention in recent years. This section reviews related works in RIS technology, its evolution into STAR-RIS, and the use of advanced optimization techniques, including machine learning and reinforcement learning, for secure and efficient multi-user communication. RIS have emerged as a promising technology for enhancing wireless communication by dynamically reconfiguring the propagation environment. RIS consists of passive, programmable meta-surfaces that manipulate the electromagnetic waves to improve signal quality, increase coverage, and reduce interference [16]. Initial research focused on optimizing beamforming and power allocation in single-user systems [17], while more recent works have extended these techniques to MIMO systems to handle the demands of modern networks [18].

Despite the success of traditional RIS, its inherent limitation to reflective-only operation restricted its ability to serve users on both sides of the surface. The introduction of STAR-RIS addressed this challenge by enabling 360-degree coverage, making it highly suitable for complex network environments [19]. STAR-RIS technology divides the surface into transmitting and reflecting regions, allowing simultaneous signal manipulation in multiple directions. Recent studies have explored its potential in optimizing spectral efficiency [20], enabling full-duplex communication [21], and integrating with millimeter-wave systems [22].

ISAC systems aim to unify wireless communication and sensing, allowing them to share resources and operate in a symbiotic manner. While traditional systems treated sensing and communication as separate entities, ISAC enables joint optimization, leading to higher spectral efficiency and reduced hardware complexity [23]. However, achieving this integration in dynamic and interference-prone environments requires advanced technologies like STAR-RIS. Studies have demonstrated the potential of RIS to improve both communication reliability and sensing accuracy in ISAC systems [24], [25].

With the proliferation of wireless devices, ensuring secure communication in RIS-assisted systems has become a critical research focus. PLS techniques, which exploit channel characteristics to safeguard against eavesdropping, have been extensively applied to RIS systems [26]. These techniques often involve optimizing beamforming and phase shifts to minimize information leakage to eavesdroppers. STAR-RIS further enhances PLS by providing greater flexibility in directing signals, enabling efficient mitigation of eavesdropping risks in multi-user scenarios [27], [28].

Optimizing the operation of RIS and STAR-RIS in complex environments is a non-convex problem that challenges traditional optimization methods. Machine learning techniques, particularly RL, have shown significant promise in addressing these challenges by learning optimal policies through interactions with the environment [29]. DR) algorithms, such as DDPG and SAC, have been successfully applied to RIS systems for tasks like beamforming optimization [13], resource allocation [30], and secure communication [31]. However, the application of DRL to STAR-RIS-assisted ISAC systems, particularly in multi-user and multi-eavesdropper scenarios, remains an underexplored area.

While prior research has made significant strides in RIS technology, STAR-RIS, and ISAC systems, existing works often focus on either communication or sensing in isolation, with limited attention to security in multi-user environments. Furthermore, the use of advanced DRL techniques for optimizing STAR-RIS-assisted ISAC systems is still in its infancy. This paper aims to bridge these gaps by proposing a comprehensive optimization framework that leverages DRL to jointly design the transmit beamforming, STAR-RIS coefficients, and receive filters for secure, efficient multi-user communication in ISAC systems.

### B. MOTIVATION AND CONTRIBUTION

The rapid evolution of wireless communication networks toward sixth-generation (6G) technologies has created an urgent demand for systems capable of integrating communication and sensing into a unified framework, termed ISAC. This integration is vital for enabling futuristic applications such as autonomous vehicles, smart cities, industrial automation, and secure IoT networks. However, ISAC systems face several significant challenges. First, the dynamic nature of wireless propagation environments, characterized by unpredictable obstacles, mobility, and interference, requires advanced technologies to optimize system performance. Second, with the proliferation of connected devices, protecting sensitive communications from eavesdroppers while maintaining high system performance has become increasingly critical. Third, designing an ISAC system that simultaneously addresses communication, sensing, and security requirements in multi-user, multi-eavesdropper scenarios involves tackling a highly non-convex and dynamic optimization problem. Traditional solutions, such as cryptographic techniques, often fall short in addressing physical-layer security needs in real-time, low-latency applications. To address these challenges, Reconfigurable Intelligent Surfaces (RIS) and their advanced counterpart, Simultaneously Transmitting and Reflecting RIS (STAR-RIS), have emerged as transformative technologies. STAR-RIS is particularly promising due to its ability to dynamically reconfigure the wireless environment, enabling 360-degree coverage and greater flexibility in beamforming. This makes it an ideal platform for developing secure and efficient ISAC systems. However, optimizing STAR-RIS-assisted ISAC systems under dynamic constraints remains an open challenge due to the inherent complexity of the problem.

This paper addresses the above challenges by proposing a novel STAR-RIS-assisted ISAC framework that employs **Deep Reinforcement Learning (DRL)** to jointly optimize the system parameters for secure, efficient multi-user communication. The key contributions of this work are:

1) **Unified System Model for STAR-RIS-Assisted ISAC:**
   - Developed a comprehensive system model incorporating STAR-RIS to enhance both communication and sensing functionalities.
   - Defined an energy-splitting constraint and orthogonality constraint to balance transmission and reflection modes while ensuring optimal performance.
2) **Formulation of the Optimization Problem:**
   - Formulated a non-convex optimization problem to maximize the average secrecy rate for multiple legitimate users while minimizing the information leakage to eavesdroppers.
   - Incorporated constraints on power, energy splitting, and SINR to ensure practical feasibility.
3) **Proposed DRL-Based Optimization Framework:**
   - Introduced two advanced DRL algorithms:
     a) **Deep Deterministic Policy Gradient (DDPG):** For deterministic policy optimization in continuous action spaces, achieving high precision in real-time system adjustments.
     b) **Soft Actor-Critic (SAC):** For probabilistic policy optimization, leveraging entropy regularization to enhance exploration and training stability.
   - Demonstrated how these algorithms adaptively optimize STAR-RIS coefficients, base station beamforming, and receiver filters in dynamic environments.
4) **Innovative Reward Mechanism:**
   - Designed a reward mechanism that balances secrecy rate maximization, echo SNR for sensing, and adherence to system constraints.
   - Integrated penalties for power and energy-splitting violations to ensure compliance with practical deployment scenarios.

This work bridges the gap between theoretical advancements in DRL and practical deployment challenges in STAR-RIS-assisted ISAC systems, setting a foundation for future research in secure, scalable 6G communication.

**TABLE 1. Comparison of RIS and STAR-RIS-assisted ISAC systems.**

| Ref. | Communication Environment | Methodology | CSI Availability | Jammer or Eve | Power Efficiency | Performance Metric |
|---|---|---|---|---|---|---|
| [16] | Single-user communication | Beamforming optimization and power allocation | Perfect | No | Energy savings with passive RIS | Improved signal quality |
| [21] | Multi-user communication | Full-duplex operation with STAR-RIS | Perfect | No | Energy-efficient dual-mode STAR-RIS | Enhanced spectral efficiency |
| [23] | ISAC systems | Sensing and communication optimization | Perfect | No | Improved sensing accuracy | Increased sensing and communication efficiency |
| [24] | Joint communication and sensing | Joint beamforming and RIS optimization | Perfect | No | Optimized STAR-RIS configuration | Maximized spectral efficiency |
| [29] | Dynamic RIS-assisted environments | Reinforcement learning for resource allocation | Partial | No | Improved energy efficiency | Optimized resource utilization |
| [31] | RIS-assisted secure communication | Deep learning for beamforming | Partial | Eve | Energy-efficient secure communication | Maximized secrecy rate |
| [28] | Secure STAR-RIS communication | Physical layer security techniques | Perfect | Eve | Flexibility in directing signals | Enhanced eavesdropping mitigation |
| Proposed Work | Multi-user, multi-eavesdropper ISAC systems | DRL-based optimization (DDPG and SAC) | Perfect | Both | Power-efficient dual-mode STAR-RIS | Maximized secrecy rate and energy efficiency |

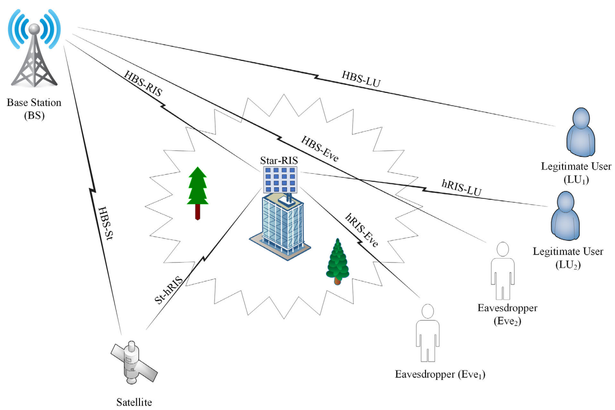



**FIGURE 1.** Secure multi-user ISAC with STAR-RIS.

## II. SYSTEM MODEL

### A. STAR-RIS MODEL

The STAR-RIS consists of $N$ programmable elements that simultaneously transmit and reflect signals. Each element is configured with amplitude and phase coefficients to optimize system performance dynamically. The transmission and reflection matrices for the STAR-RIS are given as:

$$\Theta_{\mathrm{A}} = \mathrm{diag}(\beta_{\mathrm{A},1}e^{j\theta_{\mathrm{A},1}}, \beta_{\mathrm{A},2}e^{j\theta_{\mathrm{A},2}}, \ldots, \beta_{\mathrm{A},N}e^{j\theta_{\mathrm{A},N}}), \quad (1)$$

$$\Theta_{\mathrm{R}} = \mathrm{diag}(\beta_{\mathrm{R},1}e^{j\theta_{\mathrm{R},1}}, \beta_{\mathrm{R},2}e^{j\theta_{\mathrm{R},2}}, \ldots, \beta_{\mathrm{R},N}e^{j\theta_{\mathrm{R},N}}). \quad (2)$$

Here, $\beta_{\mathrm{A},n}, \beta_{\mathrm{R},n} \in [0, 1]$ represent the amplitude coefficients for transmission and reflection, respectively, while $\theta_{\mathrm{A},n}, \theta_{\mathrm{R},n} \in [0, 2\pi)$ denote the phase shifts for the $n$-th element.

The energy-splitting constraint ensures that the total energy for each element does not exceed its capacity:

$$\beta_{\mathrm{A},n}^2 + \beta_{\mathrm{R},n}^2 = 1, \quad \forall n \in \{1, 2, \ldots, N\}. \quad (3)$$

Additionally, orthogonality between transmitted and reflected signals is maintained to reduce interference:

$$\cos(\theta_{\mathrm{R},n} - \theta_{\mathrm{A},n}) = 0, \quad \forall n \in \{1, 2, \ldots, N\}. \quad (4)$$

This configuration allows the STAR-RIS to enhance communication for legitimate users while minimizing interference for sensing and suppressing eavesdroppers.

### B. CHANNEL MODEL

The channel model captures the propagation of signals between the BS, STAR-RIS, legitimate users ($K$), and eavesdroppers ($J$) under quasi-static flat fading conditions.

#### 1) BS TO STAR-RIS CHANNEL

The channel between the BS and the STAR-RIS is modeled as Rician fading to capture the presence of a dominant Line-of-Sight (LOS) path. This choice reflects realistic propagation conditions, where a clear LOS exists due to the strategic placement of the STAR-RIS. The channel is expressed as:

$$\mathbf{H}_{\text{BS-RIS},t} = \sqrt{\lambda_{\text{BS-RIS},t}}\left(\sqrt{\frac{K}{K+1}}\mathbf{H}_{\text{LOS}} + \sqrt{\frac{1}{K+1}}\mathbf{H}_{\text{NLOS}}\right), \quad (5)$$

where:The channel between the BS and the STAR-RIS is modeled as Rician fading to account for both deterministic and random propagation effects. In this model, $\lambda_{\text{BS-RIS},t}$ represents the path loss, which accounts for large-scale fading, including distance-dependent attenuation and shadowing. The parameter $K$ is the Rician $K$factor, which quantifies the ratio of the power of the Line-of-Sight (LOS) component to the Non-Line-of-Sight (NLOS) components; a higher $K$-factor indicates a stronger LOS path. The deterministic LOS component is denoted as $\mathbf{H}_{\text{LOS}}$, which depends on the geometry and placement of the BS and STAR-RIS. The random NLOS component, $\mathbf{H}_{\text{NLOS}}$, represents scattering effects, with its entries independently drawn from a complex Gaussian distribution $\mathcal{CN}(0, 1)$.

#### 2) STAR-RIS TO USERS AND EAVESDROPPERS CHANNELS

The channel between the STAR-RIS and the $k$-th legitimate user ($\mathrm{LU}_k$) is modeled as:

$$\mathbf{h}_{\text{RIS-LU},k,t} = \sqrt{\lambda_{\text{RIS-LU},k,t}}\,\mathbf{g}_{\text{RIS-LU},k,t}, \quad (6)$$

Here, $\lambda_{\text{RIS-LU},k,t}$ represents the path loss between the STAR-RIS and the $k$-th legitimate user, accounting for large-scale fading effects such as distance-dependent attenuation and shadowing. The small-scale fading component is denoted by $\mathbf{g}_{\text{RIS-LU},k,t}$, which is modeled as a complex Gaussian random vector with entries following the distribution $\mathcal{CN}(0, 1)$, representing the random variations due to multipath propagation. $\mathcal{CN}(0, 1)$.

Similarly, the channel between the STAR-RIS and the $j$th eavesdropper (Eve$_j$) is:

$$\mathbf{h}_{\text{RIS-Eve},j,t} = \sqrt{\lambda_{\text{RIS-Eve},j,t}}\mathbf{g}_{\text{RIS-Eve},j,t}, \tag{7}$$

where:

- $\lambda_{\text{RIS-Eve},j,t}$: Path loss between the STAR-RIS and the $j$-th eavesdropper.
- $\mathbf{g}_{\text{RIS-Eve},j,t}$: Small-scale fading component, modeled as a complex Gaussian random vector $\mathcal{CN}(0, 1)$.

### 3) EFFECTIVE CHANNELS

The overall effective channel for the $k$-th legitimate user (LU$_k$) through the STAR-RIS is given by:

$$\mathbf{h}_{\text{BS-LU},k,t} = \mathbf{h}^{\top}_{\text{RIS-LU},k,t}\Theta_{\text{A,t}}\mathbf{H}_{\text{BS-RIS},t}, \tag{8}$$

while the effective channel for the $j$-th eavesdropper (Eve$_j$) is:

$$\mathbf{h}_{\text{BS-Eve},j,t} = \mathbf{h}^{\top}_{\text{RIS-Eve},j,t}\Theta_{\text{R,t}}\mathbf{H}_{\text{BS-RIS},t}. \tag{9}$$

The Rician fading model captures the presence of both deterministic LOS and random NLOS components in the BS-STAR-RIS channel, with the LOS component being dominant due to the deployment environment. On the other hand, the channels from the STAR-RIS to users and eavesdroppers are modeled with small-scale fading and path loss. This detailed modeling ensures that the proposed framework reflects realistic propagation conditions and provides a solid basis for optimizing the STAR-RIS-assisted ISAC system.

### 4) CHANNEL MODEL OVERVIEW

The channel models in this system are designed to represent realistic communication and sensing environments. These include both large-scale fading (e.g., path loss, shadowing) and small-scale fading (e.g., Rayleigh or Rician fading). The large-scale fading is captured by the path loss coefficients $\lambda_{\text{BS-RIS},t}$, $\lambda_{\text{RIS-LU},k,t}$, and $\lambda_{\text{RIS-Eve},j,t}$, while small-scale fading is modeled as a random process, where each link can be affected by multipath propagation.

For instance, the BS-to-STAR-RIS link is assumed to experience Rician fading, where the ratio of the Line-of-Sight (LOS) path to the Non-Line-of-Sight (NLOS) path is controlled by the $K$-factor. The channels between the STAR-RIS and legitimate users or eavesdroppers are modeled as Rayleigh fading for NLOS components, with the path loss for each link captured by the respective $\lambda$ values.This channel model provides the foundation for the optimization problem, where the goal is to design beamforming matrices and STAR-RIS reflection coefficients while considering the impact of different fading conditions on system performance.

## C. SIGNAL MODEL

The BS transmits a composite signal that includes both communication ($\mathbf{x}_{\text{C}}$) and sensing ($\mathbf{x}_{\text{S}}$) components:

$$\mathbf{x}_t = \mathbf{F}_{\text{C,t}}\mathbf{x}_{\text{C,t}} + \mathbf{F}_{\text{S,t}}\mathbf{x}_{\text{S,t}}, \tag{10}$$

where $\mathbf{F}_{\text{C,t}}$ and $\mathbf{F}_{\text{S,t}}$ are the beamforming matrices for communication and sensing signals.

The signal received by the $k$-th legitimate user is:

$$y_{k,t} = \mathbf{h}^{\top}_{\text{BS-LU},k,t}\mathbf{x}_t + n_{k,t}, \tag{11}$$

where $n_{k,t} \sim \mathcal{CN}(0, \sigma_k^2)$ represents additive Gaussian noise. The SINR for the $k$-th LU is:

$$\text{SINR}_{k,t} = \frac{\left|\mathbf{h}^{\top}_{\text{BS-LU},k,t}\mathbf{F}_{\text{C,t}}\mathbf{x}_{\text{C,t}}\right|^2}{\sigma_k^2 + \sum_{i\neq k}\left|\mathbf{h}^{\top}_{\text{BS-LU},k,t}\mathbf{F}_{\text{C,t}}\mathbf{x}_{\text{C,t}}\right|^2}. \tag{12}$$

The interference from the sensing signal $\mathbf{x}_{\text{S,t}}$ is omitted in Eq. (12) because the beamforming matrix $\mathbf{F}_{\text{S,t}}$ is designed to ensure orthogonality between the sensing signals and the channel responses of legitimate users, i.e., $\mathbf{h}^{\top}_{\text{BS-LU},k,t}\mathbf{F}_{\text{S,t}} \approx 0$. This ensures negligible interference from the sensing signal at legitimate users under the assumption of perfect CSI, as noted in the signal model. Hence, only intra-user interference within the communication system is considered in the SINR calculation.

The signal intercepted by the $j$-th eavesdropper is:

$$y_{j,t} = \mathbf{h}^{\top}_{\text{BS-Eve},j,t}\mathbf{x}_t + n_{j,t}, \tag{13}$$

where $n_{j,t} \sim \mathcal{CN}(0, \sigma_j^2)$. The SINR at the $j$-th Eve is:

$$\text{SINR}_{j,k,t} = \frac{\left|\mathbf{h}^{\top}_{\text{BS-Eve},j,t}\mathbf{F}_{\text{C,t}}\mathbf{x}_{\text{C,t}}\right|^2}{\sigma_j^2 + \sum_{i\neq j}\left|\mathbf{h}^{\top}_{\text{BS-Eve},j,t}\mathbf{F}_{\text{C,t}}\mathbf{x}_{\text{C,t}}\right|^2}. \tag{14}$$

The secrecy rate for the $k$-th legitimate user is defined as:

$$R^{\text{sec}}_{k,t} = \left[\log_2(1 + \text{SINR}_{k,t}) - \max_j \log_2(1 + \text{SINR}_{j,k,t})\right]^+. \tag{15}$$

The optimization goal is to maximize the average secrecy rate over $T$ time slots:

$$\max_{\Theta_{\text{A}},\Theta_{\text{R}},\mathbf{F}_{\text{C}},\mathbf{F}_{\text{S}}} \frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{K} R^{\text{sec}}_{k,t}, \tag{16}$$

subject to constraints on power, energy splitting, and SINR requirements:

$$\|\mathbf{x}_t\|^2 \leq P_{\max}, \quad \beta^2_{\text{A},n} + \beta^2_{\text{R},n} = 1. \tag{17}$$

To provide a baseline for evaluating the full potential of the proposed STAR-RIS-assisted ISAC system, this work assumes perfect CSI and orthogonal conditions for STAR-RIS elements. These assumptions allow us to focus on the core optimization problem and demonstrate the maximum achievable secrecy performance, which serves as an upper bound for practical implementations.

### D. PROBLEM FORMULATION

The objective of the problem is to optimize the joint design of the STAR-RIS transmission and reflection coefficients, as well as the beamforming matrices at the base station (BS), to maximize the secrecy performance in a multi-user, multi-eavesdropper scenario. The problem formulation considers the constraints imposed by energy splitting, SINR requirements, and transmit power limits.

The primary goal is to maximize the average secrecy rate across $T$ time slots for $K$ legitimate users ($\mathrm{LU}_k$) while mitigating the information leakage to $J$ eavesdroppers ($\mathrm{Eve}_j$). The secrecy rate for each legitimate user is defined as:

$$R_{k,t}^{\mathrm{sec}} = \left[\log_2(1+\mathrm{SINR}_{k,t}) - \max_j \log_2(1+\mathrm{SINR}_{j,k,t})\right]^+, \tag{18}$$

where: - $\mathrm{SINR}_{k,t}$ is the signal-to-interference-plus-noise ratio for $\mathrm{LU}_k$, - $\mathrm{SINR}_{j,k,t}$ is the SINR for $\mathrm{Eve}_j$ while intercepting the signal intended for $\mathrm{LU}_k$.

The objective function is to maximize the average secrecy rate:

$$\max_{\Theta_{\mathrm{A}},\Theta_{\mathrm{R}},\mathbf{F}_{\mathrm{C}},\mathbf{F}_{\mathrm{S}}} \frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{K} R_{k,t}^{\mathrm{sec}}. \tag{19}$$

The total transmit power in the BS must not exceed a predefined maximum $P_{\max}$. This is given by:

$$\|\mathbf{x}_t\|^2 \le P_{\max}, \tag{20}$$

where $\mathbf{x}_t = \mathbf{F}_{\mathrm{C,t}}\mathbf{x}_{\mathrm{C,t}} + \mathbf{F}_{\mathrm{S,t}}\mathbf{x}_{\mathrm{S,t}}$ is the transmitted signal at time $t$.

The STAR-RIS elements must balance the energy between transmission and reflection:

$$\beta_{\mathrm{A},n}^2 + \beta_{\mathrm{R},n}^2 = 1, \quad \forall n \in \{1,2,\ldots,N\}. \tag{21}$$

To ensure reliable communication for legitimate users and limit the interference of eavesdroppers, the SINR for $\mathrm{LU}_k$ must satisfy:

$$\mathrm{SINR}_{k,t} \ge \gamma_k^{\min}, \quad \forall k \in \{1,2,\ldots,K\}, \tag{22}$$

where $\gamma_k^{\min}$ is the minimum SINR threshold for $\mathrm{LU}_k$.

For sensing and suppressing eavesdroppers, the STAR-RIS design should ensure:

$$\mathrm{SINR}_{j,k,t} \le \gamma_j^{\max}, \quad \forall j \in \{1,2,\ldots,J\}, \tag{23}$$

where $\gamma_j^{\max}$ is the maximum allowable SINR at $\mathrm{Eve}_j$.

The phase shifts of the STAR-RIS elements for transmission and reflection are coupled:

$$\cos(\theta_{\mathrm{R},n} - \theta_{\mathrm{A},n}) = 0, \quad \forall n \in \{1,2,\ldots,N\}. \tag{24}$$

Combining the objective function and constraints, the optimization problem can be formulated as:

$$\begin{aligned} \max_{\Theta_{\mathrm{A}},\Theta_{\mathrm{R}},\mathbf{F}_{\mathrm{C}},\mathbf{F}_{\mathrm{S}}} \ & \frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{K}\Big[\log_2(1+\mathrm{SINR}_{k,t}) \\ & - \max_j \log_2(1+\mathrm{SINR}_{j,k,t})\Big]^+. && (25) \\ \text{subject to:} \ & \|\mathbf{x}_t\|^2 \le P_{\max}, \quad \forall t, && (25.1) \\ & \beta_{\mathrm{A},n}^2 + \beta_{\mathrm{R},n}^2 = 1, \quad \forall n, && (25.2) \\ & \mathrm{SINR}_{k,t} \ge \gamma_k^{\min}, \quad \forall k, && (25.3) \\ & \mathrm{SINR}_{j,k,t} \le \gamma_j^{\max}, \quad \forall j, && (25.4) \\ & \cos(\theta_{\mathrm{R},n} - \theta_{\mathrm{A},n}) = 0, \quad \forall n. && (25.5) \end{aligned}$$

This problem is a nonconvex optimization task due to the coupled variables and constraints, making it challenging to solve directly. Advanced techniques such as DRL are employed to address this complexity, enabling dynamic and adaptive optimization in real-time.

## III. PROPOSED DRL-BASED JOINT DESIGN ALGORITHMS

In this section, we propose two DRL-based joint design algorithms to solve the optimization problem for the STAR-RIS-assisted ISAC secure system. The joint BS beamforming and STAR-RIS phase shifts design is modeled as a single-agent Markov decision process (MDP). In the formulated MDP, the BS and STAR-RIS are modeled as a unified agent. The behavior of the agent, represented by the BS transmit beamforming matrices and the STAR-RIS transmitting and reflecting coefficients, is optimized to maximize the average secrecy rate while satisfying all system constraints.

The selection of DDPG and SAC as the core algorithms in this work is driven by their unique strengths in addressing continuous control problems, which are central to optimizing STAR-RIS phase shifts and beamforming coefficients. Unlike PPO and A3C, which are primarily designed for discrete or hybrid action spaces, DDPG and SAC excel in continuous action domains. Specifically, DDPG employs deterministic policy optimization, which allows for high precision in real-time adjustments of system parameters, a critical requirement in dynamic and adaptive STAR-RIS-assisted ISAC systems. Additionally, SAC incorporates an entropy-regularized objective function, promoting balanced exploration and exploitation, which ensures robust optimization in non-convex and dynamic environments. While PPO and A3C are known for their simplicity and stability in discrete or low-dimensional problems, they are less suited for high-dimensional, continuous optimization tasks like those presented in this work. By leveraging the advanced capabilities of DDPG and SAC, this framework achieves superior adaptability, convergence efficiency, and overall performance in maximizing secrecy rate and optimizing resource allocation for STAR-RIS-assisted ISAC systems.

The action space consists of the BS beamforming matrices $\mathbf{F}_{\mathrm{C,t}}$, $\mathbf{F}_{\mathrm{S,t}}$, and the STAR-RIS transmission and reflection coefficients $\Theta_{\mathrm{A}}$ and $\Theta_{\mathrm{R}}$. The state space includes the CSI of all involved channels, enabling the agent to dynamically adjust its actions to the current environment. Once an action $a_t$ is taken, the agent computes the reward $r_t$, which reflects the achieved secrecy rate, echo SNR, and adherence to constraints, while transitioning the environment state $s_t$ to $s_{t+1}$. To optimize the system parameters, the update for $\mathbf{u}$ reduces to a Rayleigh quotient problem, with the solution expressed as:

$$\mathbf{u}^* = \frac{\left(\mathbf{I}_{K+M} \otimes \mathbf{H}_{\text{BS-RIS},t}\right)\tilde{\mathbf{f}}}{\tilde{\mathbf{f}}^H\left(\mathbf{I}_{K+M} \otimes \mathbf{H}_{\text{BS-RIS},t}^H \mathbf{H}_{\text{BS-RIS},t}\right)\tilde{\mathbf{f}}}. \tag{26}$$

To ensure that the constraints (25.2), (25.3), and (25.4) are met, the reward mechanism is designed to penalize violations of power, energy-splitting, and SINR constraints, as shown in Algorithm 1.

**Algorithm 1** Reward Mechanism

**if** $\text{SINR}_{\min} \geq \gamma_{\min}$ for all users**and**power $\leq P_{\max}$ **then**
  **if** $R_{k,t} \leq R_{\min}$ **then**
    $r_t = \frac{\kappa_t + R_{k,t} + \text{SNR}_{\text{echo}}}{T}$
  **else**
    $r_t = \frac{\kappa_t + R_{\min} + R_{k,t}^{\text{sec}} + \text{SNR}_{\text{echo}}}{T}$
  **end if**
**else**
  $r_t = \frac{\kappa_t + \text{SNR}_{\text{echo}}}{T} - \eta_1 \cdot \text{penalty}_{\text{power}} - \eta_2 \cdot \text{penalty}_{\text{energy-split}}$
**end if**

This formulation dynamically balances system performance across secure communication, sensing, and energy constraints, ensuring compliance with the STAR-RIS design principles.

### A. THE DDPG ALGORITHM

The DDPG algorithm is a powerful method for solving continuous control problems, making it suitable for optimizing the STAR-RIS-assisted ISAC secure system. In this context, the DDPG algorithm is utilized to solve the optimization problem (25) formulated in the system model.

#### 1) KEY STEPS IN THE DDPG FRAMEWORK

The main framework of the proposed DDPG algorithm can be summarized as follows:

1) **Action Exploration with Noise:** Add exploration noise $n_0$ to the network's output to encourage exploration of the action space:

$$a_t = \pi_\varepsilon\left(s_t\right) + n_0. \tag{27}$$

2) **Soft Updates of Target Networks:** The target networks (critic and actor) are updated using soft updates from their respective primary networks:

$$\bar{\beta} \leftarrow \lambda\beta + (1-\lambda)\bar{\beta}, \quad \bar{\varepsilon} \leftarrow \lambda\varepsilon + (1-\lambda)\bar{\varepsilon}, \tag{28}$$

where $\beta, \bar{\beta}, \varepsilon$, and $\bar{\varepsilon}$ denote the critic, target critic, actor, and target actor networks, respectively, and $\lambda$ is the soft updating parameter.

3) **Critic Network Update:** The state-value function $Q(s_i, a_i \mid \beta)$ is computed using the critic network $\beta$, with a batch size $D$. The deterministic policies are updated by backpropagating through the actor network $\varepsilon$ using:

$$\nabla_\varepsilon J(\varepsilon) = \frac{1}{D}\sum_{i=1}^{D} \nabla_a Q\left(s_i, a_i \mid \beta\right) \cdot \nabla_\varepsilon \pi\left(s_i \mid \varepsilon\right). \tag{29}$$

4) **Critic Loss Minimization:** The critic network minimizes the loss function:

$$L = \frac{1}{D}\sum_{i=1}^{D}\left(y_i - Q\left(s_i, a_i \mid \beta\right)\right)^2, \tag{30}$$

where $y_i = r_i + \gamma\bar{Q}\left(\bar{s}_i, \pi_\varepsilon\left(\bar{s}_i\right) \mid \bar{\beta}\right)$ is the target state-value function computed using the target critic network, and $\gamma$ is the discount factor.

5) **Replay Buffer:** The replay buffer stores the past experiences $(s_t, a_t, r_t, \bar{s}_t)$, enabling efficient learning by sampling mini-batches for training.

#### 2) SUMMARY OF THE PROPOSED ALGORITHM

The proposed DDPG-based joint design algorithm is outlined in Algorithm 2.

#### 3) INTEGRATION WITH STAR-RIS MODEL

The DDPG algorithm dynamically adjusts:

1) **STAR-RIS Transmission and Reflection Coefficients:** $\Theta_{\mathrm{A}}$ and $\Theta_{\mathrm{R}}$ are optimized based on energy-splitting constraints (Equation (3)) and orthogonality constraints (Equation (4)).
2) **BS Beamforming Matrices:** $\mathbf{F}_{\mathrm{C}}$ and $\mathbf{F}_{\mathrm{S}}$ are updated to balance communication and sensing objectives, following the SINR constraints (Equations (25.3) and (25.4)).
3) **Objective Function:** The reward reflects the secrecy rate maximization objective (Equation (25)) while satisfying system constraints.

This DDPG-based approach ensures adaptive and efficient optimization in real-time for the STAR-RIS-assisted ISAC secure system. The use of perfect CSI in this work ensures that the DDPG and SAC algorithms can achieve optimal beamforming and resource allocation without uncertainties, providing a clear demonstration of the algorithms' effectiveness in maximizing secrecy rate and optimizing system performance. This serves as a strong benchmark for evaluating future extensions under practical conditions.

### B. THE SAC ALGORITHM

SAC is a probabilistic policy gradient algorithm based on maximum entropy theory, which adaptively adjusts the degree of exploration and enhances the learning performance.

**Algorithm 2** The Proposed DDPG-Based Joint Design Algorithm

**Input:** CSI, parameters $\lambda, \gamma, D$
**Output:** $\mathbf{f}, \Theta_{\mathrm{A}}, \Theta_{\mathrm{R}}, \mathbf{u}$
Initialize replay buffer, $\mathbf{f}$, $\Theta_{\mathrm{A}}$, $\Theta_{\mathrm{R}}$, $\mathbf{u}$, $\beta$, $\bar{\beta}$, $\varepsilon$, $\bar{\varepsilon}$
**repeat**
  Capture the initial state $s_0$
  **repeat**
    1) Compute action $a^t$ using:

$$a^t = \pi_\varepsilon\left(s^t\right) + n_0$$

    2) Observe new state $s^{t+1}$
    3) Obtain $\mathbf{u}^*$ by solving:

$$\mathbf{u}^* = \frac{\left(\mathbf{I}_{K+M} \otimes \mathbf{H}_{\text{BS-RIS},t}\right)\tilde{\mathbf{f}}}{\tilde{\mathbf{f}}^H \left(\mathbf{I}_{K+M} \otimes \mathbf{H}_{\text{BS-RIS},t}^H \mathbf{H}_{\text{BS-RIS},t}\right)\tilde{\mathbf{f}}}.$$

    4) Calculate reward $r^t$ based on Algorithm 1
    5) Store $\{a^t, s^t, r^t, s^{t+1}\}$ in replay buffer
    6) Sample random mini-batch of size $D$
    7) Update $\beta$ by minimizing the loss:

$$L = \frac{1}{D}\sum_{i=1}^{D}\left(y_i - Q\left(s_i, a_i \mid \beta\right)\right)^2$$

    8) Update $\varepsilon$ using:

$$\nabla_\varepsilon J(\varepsilon) = \frac{1}{D}\sum_{i=1}^{D} \nabla_a Q\left(s_i, a_i \mid \beta\right) \cdot \nabla_\varepsilon \pi\left(s_i \mid \varepsilon\right)$$

    9) Perform soft updates:

$$\bar{\beta} \leftarrow \lambda\beta + (1-\lambda)\bar{\beta}, \quad \bar{\varepsilon} \leftarrow \lambda\varepsilon + (1-\lambda)\bar{\varepsilon}$$

    every $G$ steps
    10) Update $t \leftarrow t+1$
  **until** convergence
**until** convergence

Additionally, SAC employs two critic networks to stabilize the training process.

#### 1) ALGORITHM SUMMARY

The proposed SAC-based joint design algorithm is summarized in Algorithm 3.

#### 2) SAC OBJECTIVE

The DDPG algorithm aims to maximize the expected accumulated reward to find the optimal policy:

$$\pi^* = \arg\max_\pi \mathbb{E}_{(s_t, a_t)\sim\rho_\pi}\left[\sum_t r\left(s_t, a_t\right)\right]. \tag{31}$$

In contrast, SAC also incorporates the maximum entropy objective, enabling stochastic policy selection. This is expressed as:

$$\pi^* = \arg\max_\pi \mathbb{E}_{(s_t, a_t)\sim\rho_\pi}\left[\sum_t r\left(s_t, a_t\right) + \alpha\mathcal{H}\left(\pi\left(\cdot \mid s_t\right)\right)\right], \tag{32}$$

where $\alpha$ is the temperature parameter, and $\mathcal{H}\left(\pi\left(\cdot \mid s_t\right)\right)$ is the entropy of the policy.

**Algorithm 3** The Proposed SAC-Based Joint Design Algorithm

**Initialization:** Replay buffer, network parameters $\omega, \mu_1, \mu_2, \psi$
**for** each episode **do**
  Capture the beginning state $s_0$
  **for** each environment step **do**
    Obtain $a_t \sim \pi_w\left(a_t \mid s_t\right)$
    Observe $s_{t+1} \sim p\left(s_{t+1} \mid s_t, a_t\right)$
    Obtain $\mathbf{u}^*$ by solving:

$$\mathbf{u}^* = \frac{\left(\mathbf{I}_{K+M} \otimes \mathbf{H}_{\text{BS-RIS},t}\right)\tilde{\mathbf{f}}}{\tilde{\mathbf{f}}^H \left(\mathbf{I}_{K+M} \otimes \mathbf{H}_{\text{BS-RIS},t}^H \mathbf{H}_{\text{BS-RIS},t}\right)\tilde{\mathbf{f}}}.$$

    Calculate reward $r^t$ based on Algorithm 1
    Store $\{a^t, s^t, r^t, s^{t+1}\}$ in the replay buffer
  **end for**
  **for** each gradient step **do**
    Sample random mini-batch $D$
    Update $\mu_1, \mu_2$ using:

$$\mu_j \leftarrow \mu_j - \lambda\nabla_{\mu_j} J_Q\left(\mu_j\right), \quad j \in \{1, 2\}$$

    Update $\omega$ using:

$$\omega \leftarrow \omega - \lambda_\omega \hat{\nabla}_w J_\pi(\omega)$$

    Update $\psi$ using:

$$\psi \leftarrow \psi - \lambda_\psi \hat{\nabla}_\psi J_V(\psi)$$

    Perform soft updates for target networks:

$$\bar{\mu} \leftarrow \lambda\mu + (1-\lambda)\bar{\mu}, \quad \bar{\psi} \leftarrow \lambda\psi + (1-\lambda)\bar{\psi}$$

  **end for**
**end for**

#### 3) POLICY EVALUATION

The state-action value function is defined as:

$$Q\left(s_t, a_t\right) = r\left(s_t, a_t\right) + \gamma\mathbb{E}_{s_{t+1}\sim p}\left[V\left(s_{t+1}\right)\right], \tag{33}$$

where the value function $V(s_{t+1})$ is given by:

$$V(s_{t+1}) = \mathbb{E}_{a_{t+1}\sim\pi}\left[Q(s_{t+1}, a_{t+1}) - \alpha\log\pi\left(a_{t+1} \mid s_{t+1}\right)\right]. \tag{34}$$

### 4) POLICY IMPROVEMENT

The policy is updated by minimizing the Kullback-Leibler (KL) divergence:

$$\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} D_{\text{KL}} \left( \pi' \left( \cdot \mid s_t \right) \parallel \frac{\exp\left(Q^{\pi_{\text{old}}}\left(s_t, \cdot\right)\right)}{Z^{\text{old}}\left(s_t\right)} \right), \tag{35}$$

where $Z^{\text{old}}(s_t)$ is the normalization factor.

### 5) NETWORK UPDATES

SAC employs three types of networks:

- $\pi_w(a_t \mid s_t)$: Policy network
- $V_\psi(s_t)$: Value network
- $Q_\mu(s_t, a_t)$: Q-network

### 6) GRADIENT UPDATES

- Q-network:

$$\begin{aligned}\hat{\nabla}_\mu J_Q(\mu) = \nabla_\mu Q_\mu\left(a_t, s_t\right) \big( Q_\mu\left(s_t, a_t\right) \\ - r\left(s_t, a_t\right) - \gamma V_{\bar{\psi}}(s_{t+1}) \big).\end{aligned} \tag{36}$$

where $V_{\bar{\psi}}(s_{t+1})$ is the target value network.

- Policy network:

$$\begin{aligned}\hat{\nabla}_w J_\pi(w) = \nabla_w \log \pi_w\left(a_t \mid s_t\right) \\ + \Delta_{a_t} \nabla_w f_w\left(\epsilon_t; s_t\right).\end{aligned} \tag{37}$$

where $\Delta_{a_t} = \nabla_{a_t} \log \pi_w\left(a_t \mid s_t\right) - \nabla_{a_t} Q(s_t, a_t)$.

- Value network:

$$\begin{aligned}\hat{\nabla}_\psi J_V(\psi) = \nabla_\psi V_\psi(s_t) \big( V_\psi(s_t) \\ - Q_\mu(s_t, a_t) + \log \pi_w\left(a_t \mid s_t\right) \big).\end{aligned} \tag{38}$$

Thus, the proposed SAC-based joint design algorithm is summarized in Algorithm 3. The proposed DRL algorithms (DDPG and SAC) utilize neural networks with a multi-layer perceptron (MLP) architecture for both the actor and critic networks in DDPG, and the policy, value, and Q-networks in SAC. Each network consists of two hidden layers with 128 neurons, employing ReLU activation functions. The forward pass through these networks has a time complexity of $O(N)$, where $N$ is the number of neurons in the hidden layers, while the backpropagation process, required for weight updates, has a time complexity of $O(N^2)$. As the number of users, RIS elements, or other system parameters increases, the complexity of both the state and action spaces grows linearly, increasing computational demands. However, the use of experience replay, mini-batch learning, and soft target updates helps mitigate this growth, making the algorithms scalable to larger systems. In practice, the computational complexity of training both algorithms is dominated by the neural network forward passes and backpropagation, with overall complexity scaling with the number of episodes, batch size, and system size.

**TABLE 2. Simulation parameters for STAR-RIS-assisted ISAC system.**

| Parameter | Value |
|---|---|
| Number of STAR-RIS elements ($N$) | 128 (Assumed, update as needed) |
| Number of legitimate users ($K$) | 5 (Figure 6) |
| Number of eavesdroppers ($J$) | 3 (Figure 6) |
| Energy-splitting ratios | $\{0.5, 0.7\}$ (Figure 2, 3) |
| Rician $K$-factor | 10 dB (Assumed, update if needed) |
| Path loss exponent | 2.2 (Assumed, typical value) |
| Maximum transmit power ($P_{\max}$) | 30 dBm (Figure 7, typical) |
| Noise power ($\sigma^2$) | -90 dBm (Assumed, typical value) |
| Beamforming algorithms | DDPG, SAC (Figures 4a, 4b, 4c) |
| Fading scenarios | Rician, Rayleigh (Figure 6) |
| Simulation episodes | 2000 (Figure 2) |
| SINR thresholds for LUs ($\gamma_k^{\min}$) | 10 dB (Implied, confirm) |
| SINR thresholds for Eves ($\gamma_j^{\max}$) | 0 dB (Implied, confirm) |

## IV. SIMULATION RESULTS

Figure 2 illustrates the impact of varying the Energy-Splitting Ratio on the performance of a STAR-RIS-assisted ISAC system, highlighting the interplay between sensing, communication, and security. As the energy-splitting ratio increases, the Average Secrecy Rate consistently improves, demonstrating enhanced physical-layer security by minimizing data leakage to eavesdroppers. Similarly, the SINR for Communication increases, indicating better communication quality due to the higher energy allocation for transmission. However, the SNR for Sensing improves at a slower rate and eventually plateaus, as less energy is available for sensing. This behavior reveals an inherent trade-off: prioritizing energy for communication and security comes at the cost of sensing performance. Thus, Figure 3 emphasizes the importance of carefully optimizing the energy split ratio to effectively balance security, communication, and sensing requirements in ISAC systems supported by STAR-RIS.

In Figure 3, the average secrecy rate for 2000 episodes is compared for different configurations of the STAR-RIS. Three scenarios are examined: first, with STAR-RIS (0.7 Split), where the energy is split with 0.7 allocated to transmission and 0.3 to reflection. This configuration achieves the highest average secrecy rate, highlighting the significant performance improvement provided by STAR-RIS when optimized for resource allocation. The second scenario, with STAR-RIS (0.5 Split), uses an equal energy split of 0.5 for both transmission and reflection, resulting in a moderate improvement in secrecy rate compared to the baseline. Finally, the third scenario, without STAR-RIS, shows the traditional system without any STAR-RIS enhancement, resulting in the lowest average secrecy rate due to the absence of reflective beamforming enhancements. The figure demonstrates that incorporating STAR-RIS with an optimized energy split leads to notable improvements in the secrecy rate, outperforming the conventional system without STAR-RIS. Figures 4a, 4b, and 4c collectively demonstrate the performance comparison between the DDPG and SAC algorithms for the STAR-RIS-assisted ISAC system. Figure 4a shows that SAC consistently achieves higher computation efficiency (measured in steps/second)

**TABLE 3. Comparison between DDPG and SAC algorithms for STAR-RIS-assisted ISAC system.**

| Feature | DDPG | SAC |
|---|---|---|
| **Action Space** | Continuous | Continuous |
| **Policy Type** | Deterministic | Stochastic |
| **Exploration Strategy** | Noise added to actions for exploration (Ornstein-Uhlenbeck process) | Entropy maximization encourages exploration |
| **Critic Networks** | 1 (Q-function) | 2 (Q-function and Value function) |
| **Soft Updates of Target Networks** | Yes | Yes |
| **Entropy Regularization** | No | Yes, enhances exploration |
| **Convergence Speed** | Slower due to deterministic policy | Faster convergence due to entropy regularization |
| **Learning Stability** | Prone to instability in complex environments | More stable due to double Q-learning and entropy term |
| **Reward Maximization** | Direct reward maximization | Reward maximization + entropy maximization |
| **Exploration vs. Exploitation Balance** | Hand-tuned via exploration noise | Automatically balanced through entropy regularization |

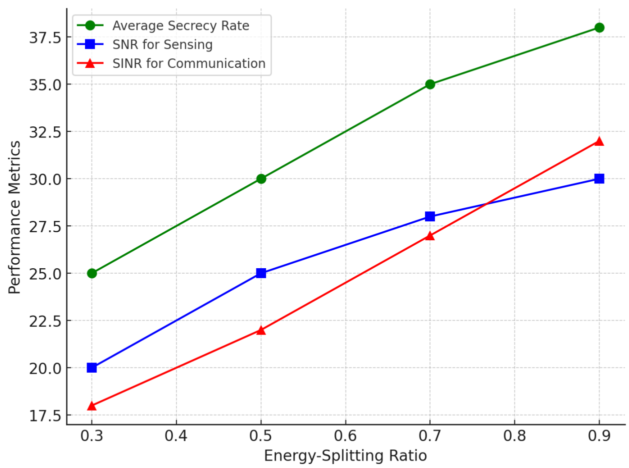



**FIGURE 2. Comparison of average secrecy rate across episodes for STAR-RIS configurations.**

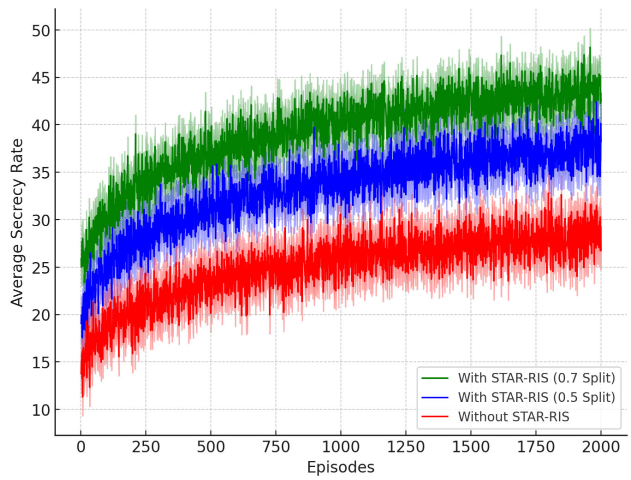
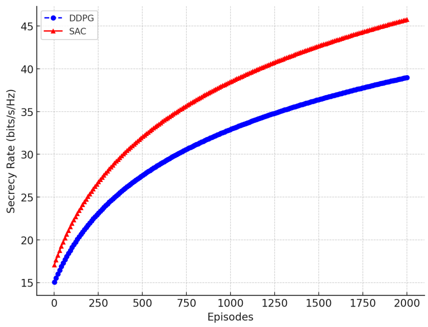



**FIGURE 3. Performance trade-offs in STAR-RIS-assisted ISAC with varying energy-splitting ratios.**

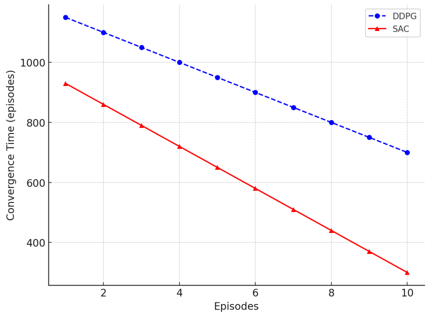
compared to DDPG, reflecting SAC's faster processing capabilities as the number of episodes increases. In Figure 4b, SAC outperforms DDPG in terms of convergence time, with SAC requiring fewer episodes to converge to optimal performance, highlighting its superior learning efficiency and faster adaptation to the dynamic optimization problem. Finally, Figure 4c illustrates that SAC achieves a higher secrecy rate (measured in bits/s/Hz) throughout the training process, indicating its effectiveness in optimizing secure communication for the ISAC system. Together, these results demonstrate that SAC provides significant improvements over DDPG in terms of computational efficiency, faster convergence, and enhanced secrecy rate, making it a more suitable algorithm for real-time and secure optimization in STAR-RIS-assisted ISAC systems.

Figure 5 demonstrates the SINR performance for legitimate users and eavesdroppers over a series of episodes in the proposed STAR-RIS-assisted ISAC system. The blue line represents the SINR for legitimate users, which consistently increases as the episodes progress, indicating improved communication quality and security. In contrast, the red line shows the SINR for eavesdroppers, which remains significantly lower, highlighting the system's effectiveness in suppressing eavesdropping capabilities. The gap between the SINR values of legitimate users and eavesdroppers widens over time, ensuring robust physical-layer security while maintaining high-quality communication for legitimate users. This result aligns with the goal of maximizing the average secrecy rate while mitigating the interference and information leakage caused by eavesdroppers.

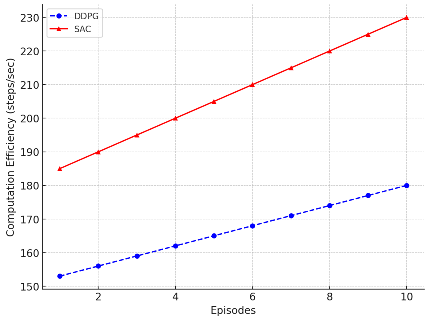
The figure 6 illustrates the secrecy rate (dB) performance as a function of the number of users in the proposed STAR-RIS-assisted ISAC system. The plot compares the secrecy rates for legitimate users (LUs) and eavesdroppers (Eves) under two different fading scenarios: Rician fading and Rayleigh fading. As the number of users increases, the secrecy rate decreases for all cases due to growing interference and limited available resources. The legitimate users consistently achieve higher secrecy rates than the eavesdroppers, showcasing the system's ability to enhance




(a) Secrecy rate (bits/s/Hz) over episodes




(b) Convergence time (episodes)




(c) Computation efficiency (steps/second)

**FIGURE 4.** **Performance comparison of DDPG and SAC algorithms.**

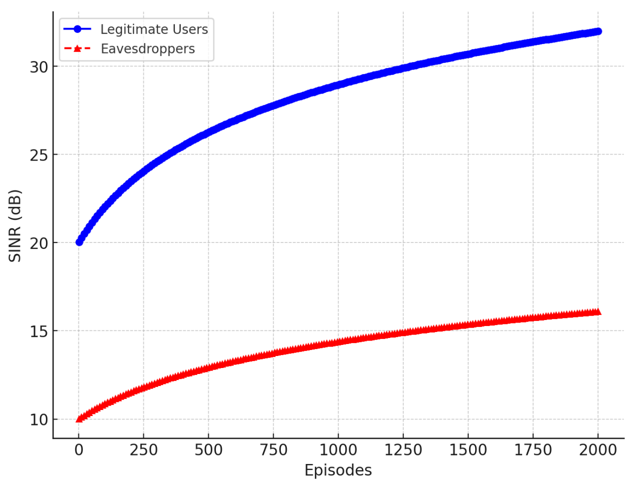



**FIGURE 5.** **SINR performance for legitimate users and eavesdroppers over episodes in the STAR-RIS-assisted ISAC system.**

physical-layer security. Moreover, the Rician fading scenario achieves higher secrecy rates compared to Rayleigh fading, highlighting the positive impact of a line-of-sight (LoS)

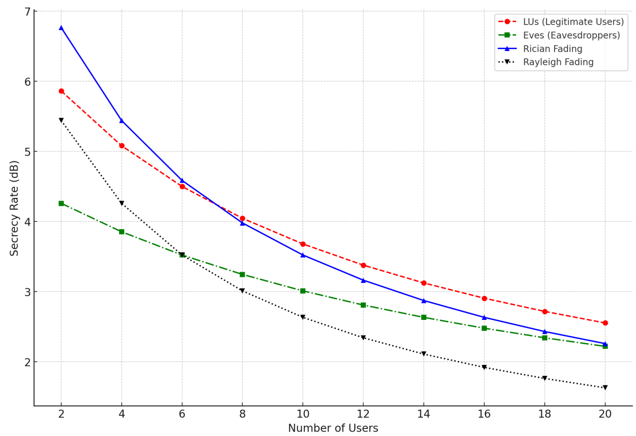



**FIGURE 6.** **Secrecy rate vs. number of users for legitimate users and eavesdroppers under Rician and Rayleigh fading in the STAR-RIS-assisted ISAC system.**

component on overall system performance. This figure demonstrates the effectiveness of the proposed framework in maintaining secure communication for legitimate users while suppressing information leakage to eavesdroppers, even as the number of users increases.

Figure 7 presents the secrecy rate (dB) performance as a function of power consumption for the proposed STAR-RIS-assisted ISAC system compared to a traditional RIS system. As power consumption increases, both systems experience an improvement in secrecy rate. However, the STAR-RIS-assisted ISAC system consistently outperforms the traditional RIS system by achieving higher secrecy rates across all power levels. The enhanced performance of the ISAC system assisted with STAR-RIS is attributed to its ability to transmit and reflect signals simultaneously, providing greater flexibility in beamforming and optimizing system resources. This demonstrates the superiority of STAR-RIS in enhancing the security of the physical layer while efficiently utilizing the available power.

SAC outperforms DDPG in the STAR-RIS-assisted ISAC problem due to its entropy regularization, stochastic policy

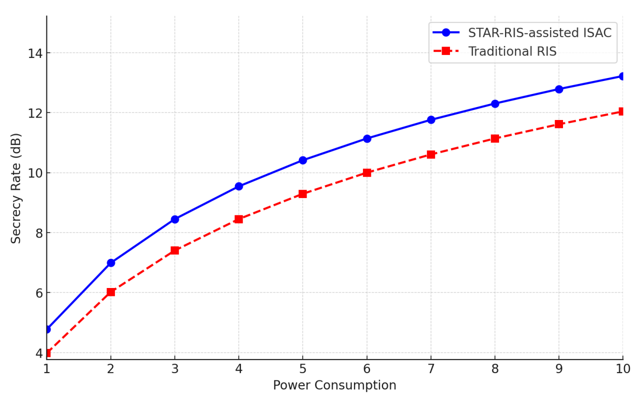



**FIGURE 7.** **Secrecy rate performance versus power consumption for STAR-RIS-assisted ISAC and traditional RIS systems.**

framework, faster convergence, and superior secrecy rate optimization. While DDPG's deterministic policies are more prone to instability, it is included as a benchmark to ensure transparency and fairness. Though SAC is optimal for this scenario, DDPG may be preferable in simpler, lower-dimensional settings where computational efficiency and stability are prioritized. This study provides a comprehensive comparison, highlighting SAC's robustness while acknowledging DDPG's potential use cases, laying the foundation for future research in ISAC and reinforcement learning.

## V. CONCLUSION

This paper proposed a STAR-RIS-assisted ISAC system to enhance secure communication and sensing in multi-user environments with multiple eavesdroppers. By jointly optimizing the base station's beamforming, STAR-RIS transmission and reflection coefficients, and receive filters, the system maximized the long-term average secrecy rate while meeting sensing SNR and achievable rate constraints. To address the complexity of the non-convex optimization, two DRL algorithms, DDPG and SAC, were employed for adaptive and efficient optimization.

Numerical results demonstrated that the proposed system significantly outperforms conventional RIS setups, achieving higher secrecy rates, faster convergence, and better computational efficiency. This work showcases STAR-RIS as a promising technology for secure and integrated communication and sensing, offering scalable solutions for 6G networks, smart cities, and IoT applications.

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